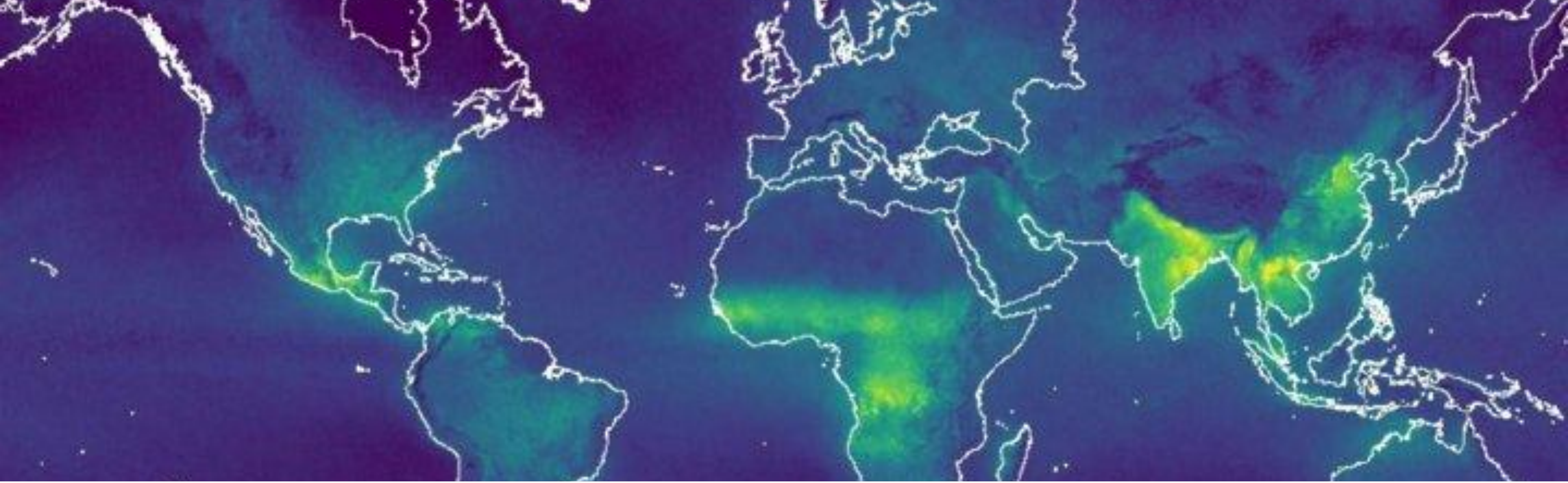
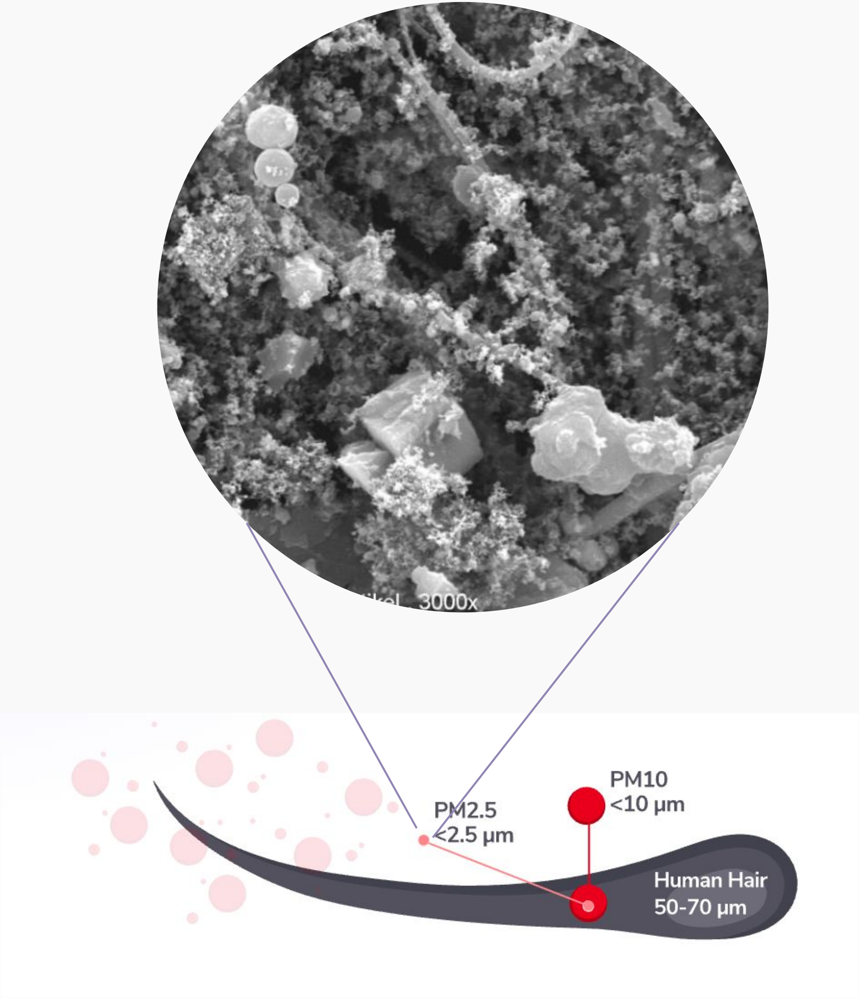


Urban Air Pollution Machine Learning Project

Helge Linnert

January 2026

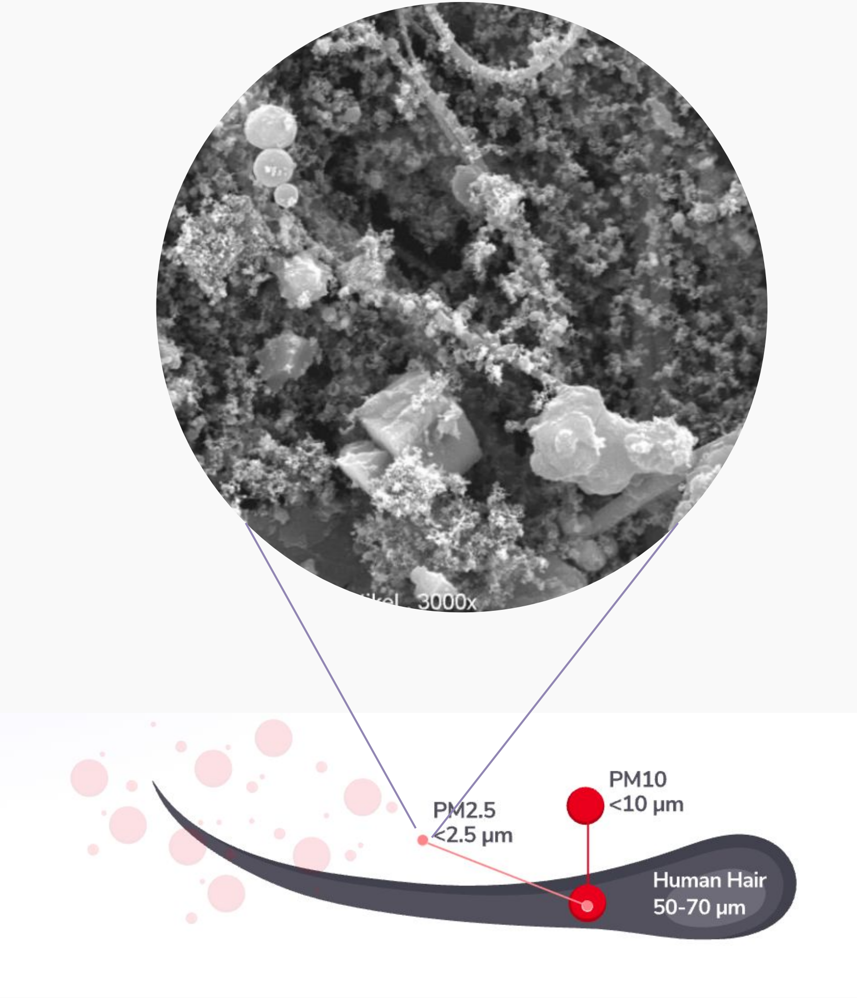
Particulate Matter
(PM2.5)



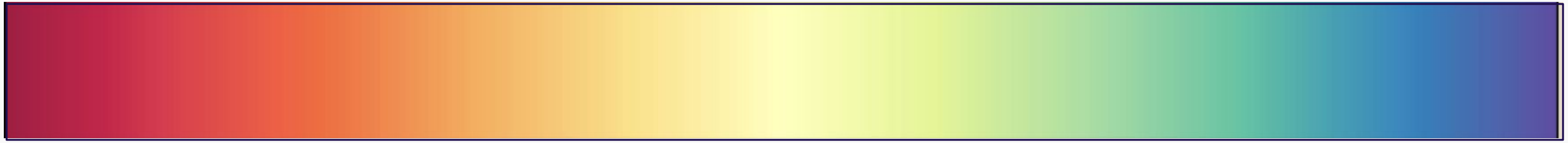
Particulate Matter (PM2.5)

→ Linked to cardiovascular diseases

→ Reduces life-expectancy



WHO Air Quality Index

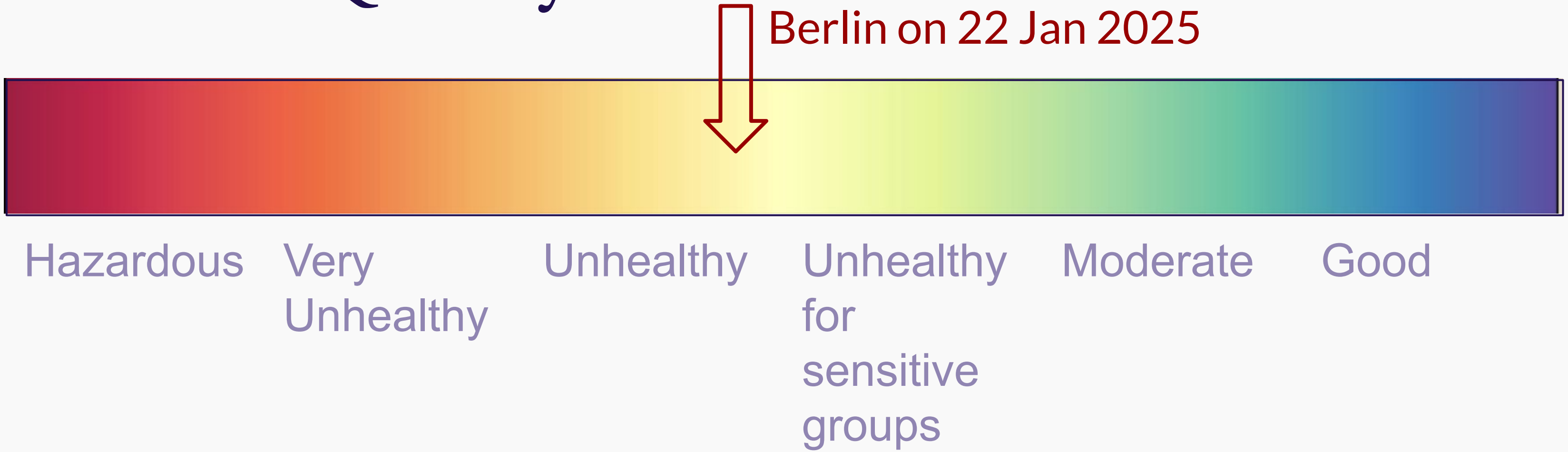


Hazardous | Very Unhealthy | Unhealthy | Unhealthy for sensitive groups | Moderate | Good

Good: 0–12 μg/m^3
Moderate: 12.1–35.4 μg/m^3
Unhealthy for sensitive groups: 35.5–55.4 μg/m^3
Unhealthy: 55.5–150.4 μg/m^3
Very unhealthy: 150.5–250.4 μg/m^3
Hazardous: >250.5 μg/m^3

LINK

WHO Air Quality Index



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Hazardous: >250.5 µg/m^3

LINK

“How high is the concentration of particulate matter (PM2.5) in the air?”

- **Motivation**
- **Data Set**
- **Predictions**
- **Next Steps**

Motivation

→ Forecast air pollution

→ Protect population health

→ Inform decision-making of policies

Data Set

- → Satellite and weather data
- → 340 locations around the globe
- → 1st Feb - 4th Apr 2020
- → 30,557 entries, 82 different properties

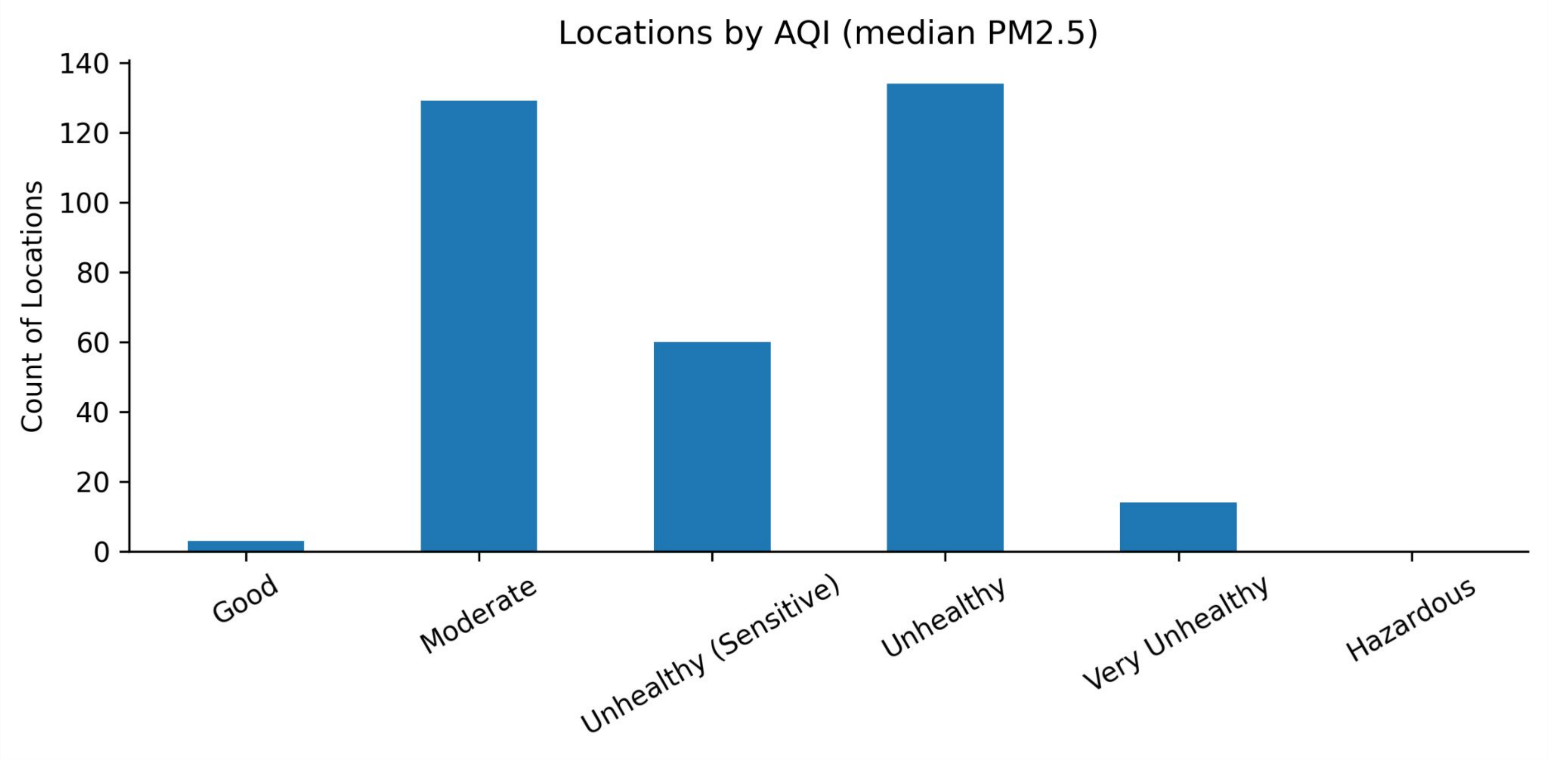
Data Source

→ Google Earth Engine Data Catalog

→ In cooperation with

Link to Satellite and Weather data

Data Set


Locations by AQI (median PM2.5)
Count of Locations
0
20
40
60
80
100
120
140
Good
Moderate
Unhealthy (Sensitive)
Unhealthy
Very Unhealthy
Hazardous

Data Set - Variables

- → Wind speed (m/s), temperature (°C), relative humidity, ...
- → Trace gases (NO_2, O_3, CO, HCHO, SO_2, CH_4)
- → Properties: Solar & sensor angles, column density, ...
- → e.g. L3_SO2_cloud_fraction

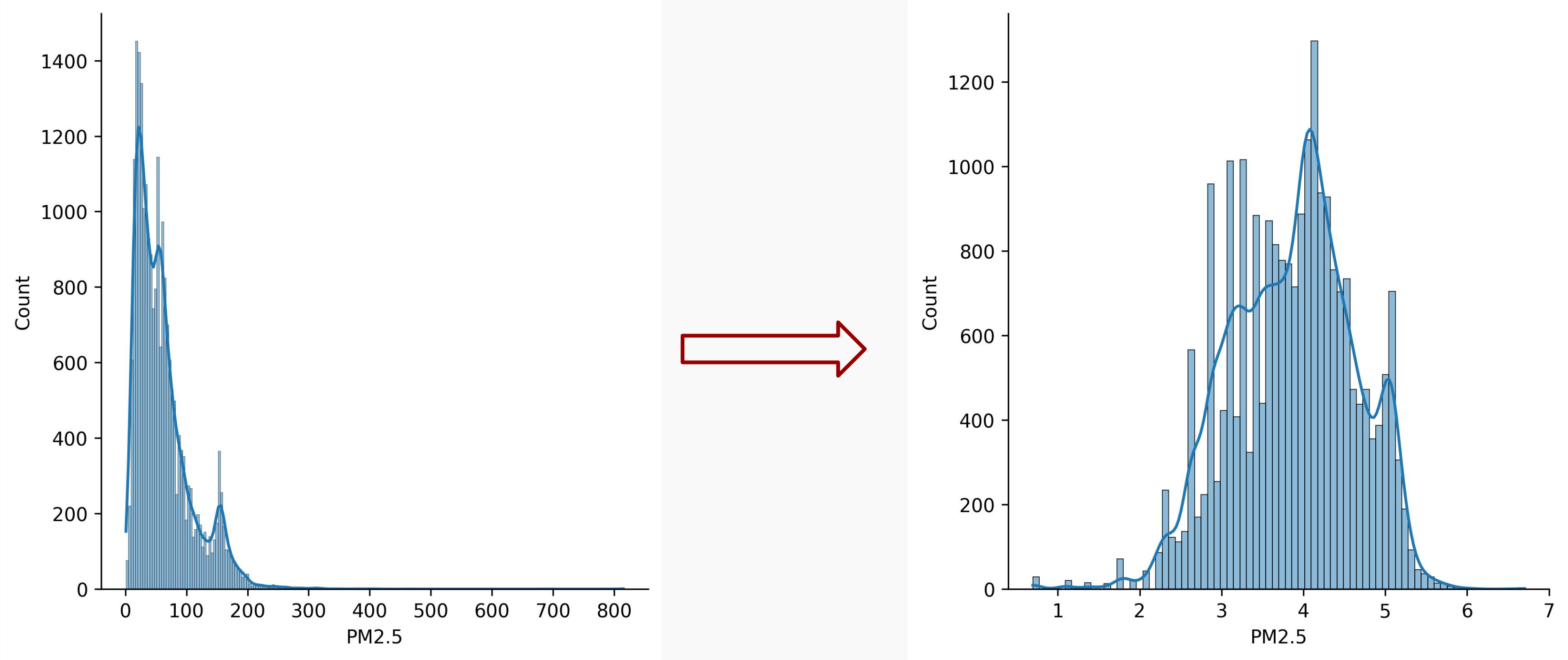
Data Set - Challenges

→ Highly technical data

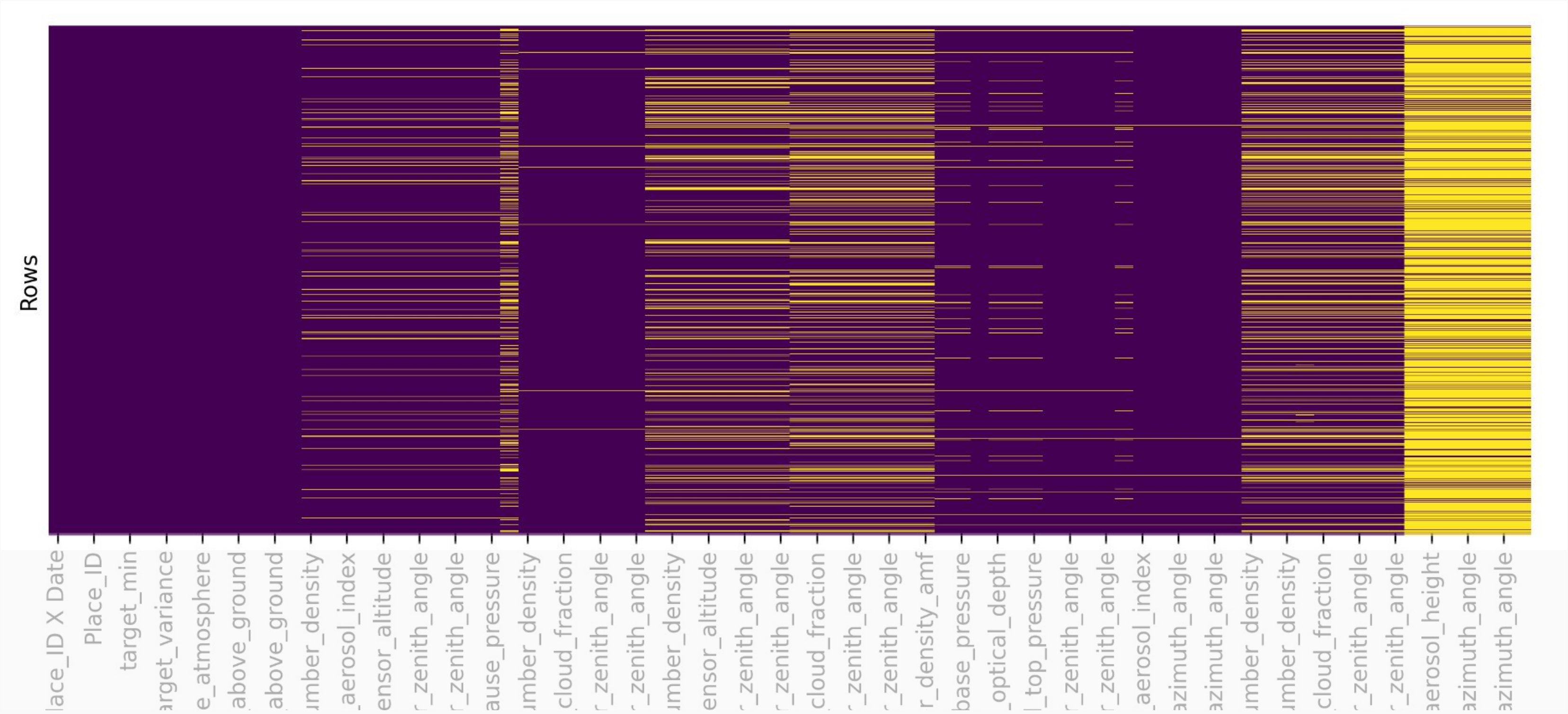
→ Imbalanced, missing values, outliers

→ Fluctuations due to covid-19 influences

Data Set - Challenges



Data Set - Challenges



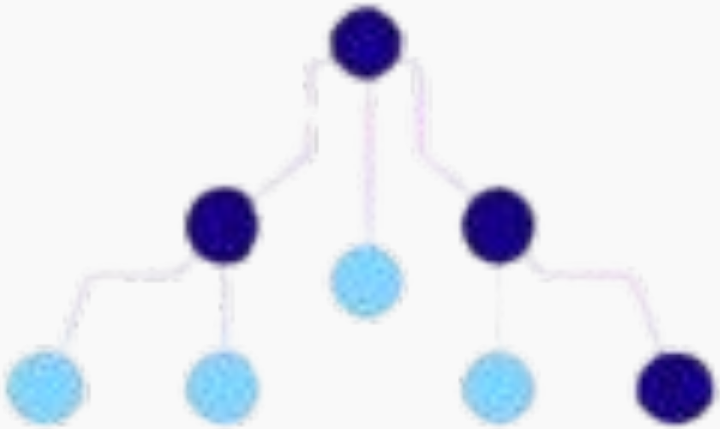
Predictions

- → Baseline: 46.47 μg/m^3
- → Advanced: 28.68 μg/m^3
- → Performance: how far off from reality on average (RMSE)

Predictions - Factors

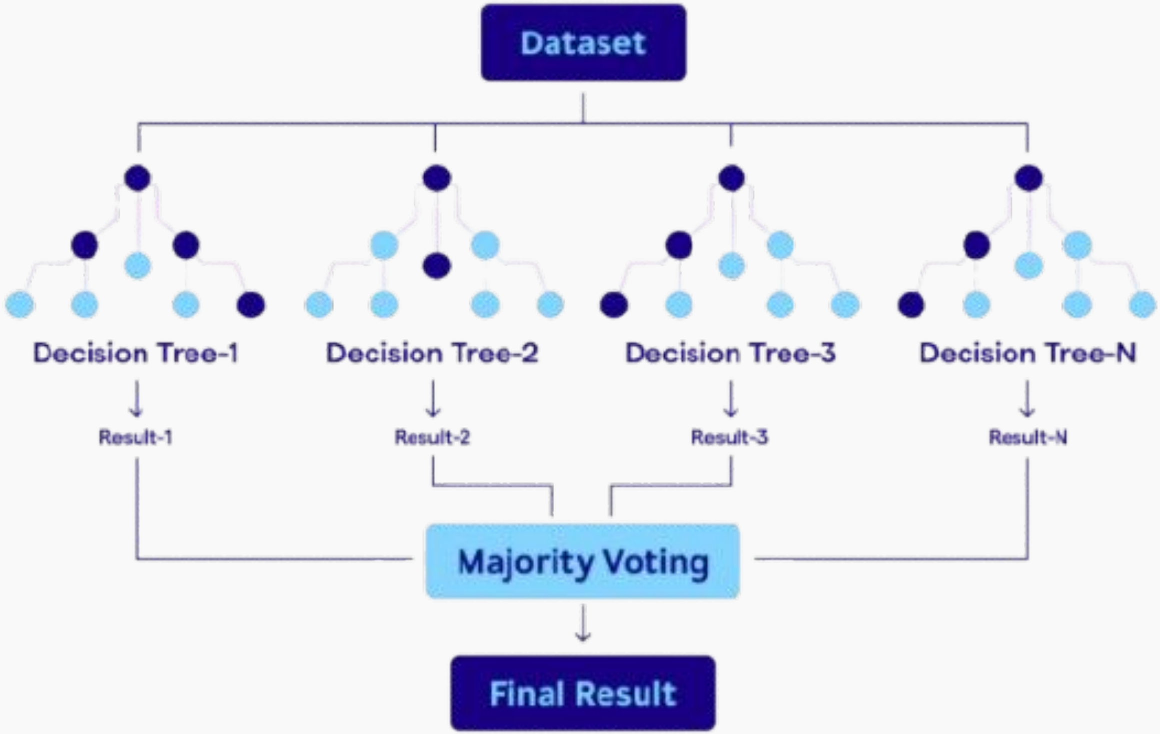
- → Top: Carbon Monoxide, Formaldehyde and Nitrogen Dioxide, … (top 10 correlations)
- → Low: Methane, Ozone, Sulphur Dioxide
- → Best Strategy: Decision Tree (RandomForest)

Predictions

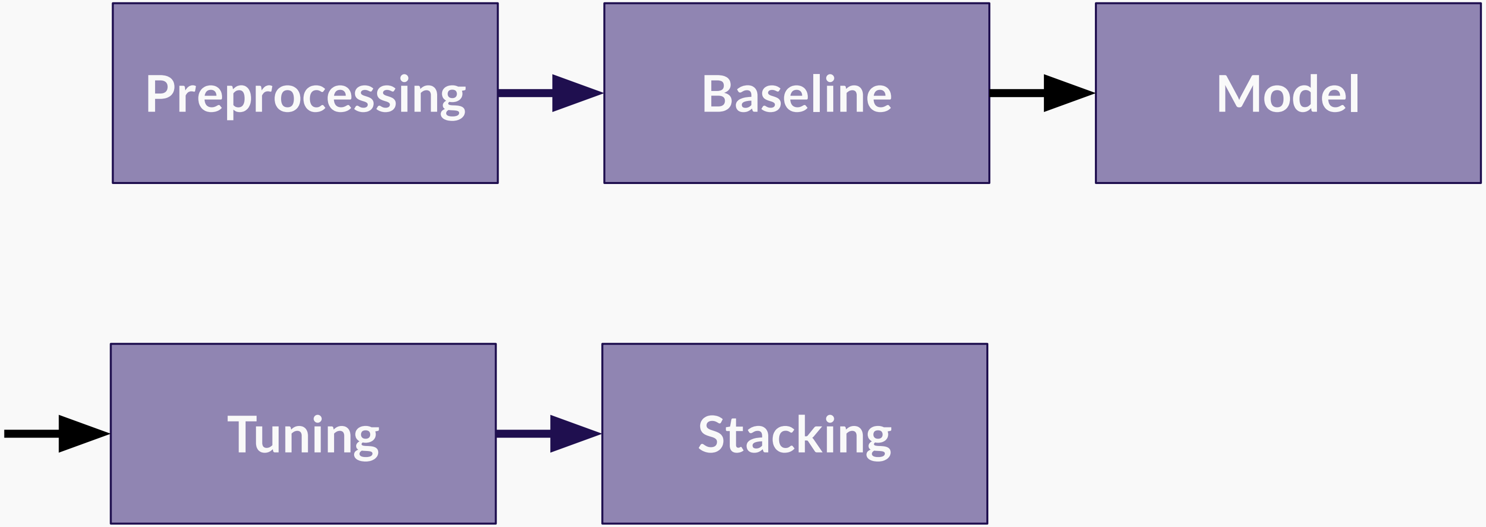


Decision Tree-1

Predictions



ML Workflow



Next Steps

→ Scrutinize data regarding outliers, weekday, re covid-19

→ Improve predictions: many trees to learn from mistakes of previous models (xgboost)


Thank you !